

Troy Mayer

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EDUCATION

Purdue University, School of Mechanical Engineering and School of Science <i>B.S. Mechanical Engineering; B.S. Applied Statistics; Minor in Economics</i> <i>Clubs: Club Swimming, House Band</i>	West Lafayette, IN May 2028 GPA:3.4/4
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PROFESSIONAL EXPERIENCE

Boston Dynamics <i>Sensors and Perception Co-Op – Atlas</i>	Waltham, MA July 2026 – Present
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- Characterized hundreds of impact events by decomposing force vectors to localize load application, applying Butterworth filtering and fitting damped sinusoidal models to extract natural frequency, damping ratio, and absorbed energy.
- Modeled hardware thermal behavior in Python (Pandas, SciPy), applying Savitzky-Golay smoothing and regime detection within heating cycles to fit exponential segments via multi-start least squares; projected steady-state temperatures across hundreds of units, gated on residual RMSE.
- Built and deployed a Flask/Plotly dashboard that runs the fitting pipeline automatically as test runs complete, with drill-down into individual time series and fit diagnostics; deployed on a Linux VM for continuous use by engineers during live testing.
- Extended onboard logging to capture additional impact and telemetry data during live operation, expanding the dataset from controlled tests to real-world runs.

Caterpillar (via Purdue Data Mine) <i>Machine Learning Researcher</i>	West Lafayette, IN January 2026 – May 2026
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- Built an LSTM time-series forecasting model to predict material consumption rates from historical sequence data, achieving 5% reduction in material usage.
- Developed Monte Carlo simulations over thousands of iterations to quantify variance in resource allocation outcomes under uncertainty.
- Built Python pipelines to ingest, clean, and structure months of noisy legacy data, contributing to a 4% reduction in manufacturing cycle time.

Chortos Lab – Soft Adaptive Systems and Structures <i>Undergraduate Researcher</i>	West Lafayette, IN August 2025 – May 2026
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- Built data acquisition systems in LabVIEW and MATLAB, processing and normalizing datasets of 50,000+ points to quantify sub-millimeter material behavior.

PROJECTS

Purdue ASME – Biomechanics <i>Active Orthosis – Electronics Team Lead</i>	West Lafayette, IN April 2024 – Present
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- Wrote real-time C++ firmware closing a 500 Hz control loop from EMG sensor input to actuator command, holding end-to-end latency under 10 ms.
- Designed and routed a custom PCB control hub interfacing driver signal with sensor input as the system's central processing unit.
- Led a 5-engineer team integrating FEA-validated structural design into a chassis producing over 150 N of force.

Legged Robot State Estimation

<i>Personal Project</i>	August 2026 – Present
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- Implemented a contact-aided error-state EKF in C++/Eigen estimating floating-base pose and velocity for a simulated quadruped, fusing IMU propagation with leg-kinematics updates; bounded velocity RMSE to 0.08 m/s against ground truth over a 60-second trajectory.
- Characterized observability under IMU and kinematics alone, showing roll, pitch, and velocity remain bounded while position and yaw drift without limit; verified filter Jacobians against finite differences in a GoogleTest suite, with the estimator running at 1 kHz.

SKILLS & INTERESTS

- **Tools:** C++, Python, MATLAB, SQL, LabVIEW, CAD (NX, Fusion, Creo), FEA Analysis, KiCad, Snowflake, Git, Linux, Eigen, MuJoCo
- **Languages:** English (Native), Turkish (Fluent), Spanish (Fluent - New Jersey Seal of Biliteracy).
- **Hobbies:** Lifting, Hiking, Guitar, Saxophone